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Abstract

Hamiltonian strong coupling expansions for the string tension and β-functions of SU(3) lattice gauge theory are presented to $O(g^{-24})$ in 3 + 1 and 2 + 1 dimensions. The 3 + 1 dimensional theory is seen to have a weak coupling region $0 \leq g \leq 1.05$, a narrow crossover region, $1.05 \leq g \leq 1.50$, and a strong coupling region, $g \geq 1.50$. These features follow from the series without the use of extrapolation methods. The numerical relation between the string tension T and Λ^{mom} is obtained, $\sqrt{T} = (2.50 \pm .50)\ \Lambda^{mom}$ in good agreement with recent computer simulations. The width of the flux tube is computed to $O(g^{-20})$ and we find evidence for its divergence ("roughening") at a coupling $g \lesssim 1.02$, inside the <u>weak</u> coupling regime.

Numerical studies of lattice gauge theories have recently shed considerable light on the physics of confinement. These studies include Hamiltonian[1], Euclidean strong coupling expansions[2] and Euclidean computer simulations using Monte Carlo methods.[3] Some of the most fruitful work concerns the dependence of the theory's string tension on its coupling constant. In the Hamiltonian formulation of the problem one studies the energy per unit length of the flux tube connecting a static quark and an anti-quark in the theory of only gluons. The Hamiltonian of the theory is,

$$H = \frac{g^2}{2a}\left(\sum_{\ell} E_{\ell}^2 - x \sum_{p} \mathrm{tr}\ [U(p) + \mathrm{h.c.}]\right) \quad , \qquad x = \frac{2}{g^4} \tag{1}$$

where E_{ℓ}^2 is the quadratic Casimir operator of SU(3) on the link ℓ of a three dimensional cubic lattice and U(p) is the unitary fundamental representation of the product of group elements on the boundary of a plaquette p. The static quark and anti-quark are connected by a chromo-electric flux tube and the energy per unit length ("string tension") can be computed as a power series in $x = 2/g^4$,

$$T = \frac{g^2}{2a^2} \cdot W(x) \tag{2a}$$

where

$$W(x) = \sum_{n=0}^{\infty} t_n x^n \tag{2b}$$

As we have discussed elsewhere[1] the string tension can be used to renormalize the lattice theory -- one requires that T be fixed, independent of "a", the lattice spacing. Then g, the lattice coupling constant, must depend upon "a" in a well-defined fashion and an expression for the theory's Callan-Symanzik β-function follows[1],

$$\frac{\beta(g)}{g} = - \frac{d\ln g}{d\ln a} = \frac{-1}{1-2xW'/W} \tag{3}$$

The central question of quark confinement is addressed by using Eq. (3). For g near zero weak coupling perturbation theory gives,

$$- \frac{\beta(g)}{g} = \frac{11}{16\pi^2} g^2 + \frac{102}{(16\pi^2)^2} g^4 + \ldots \tag{4a}$$

while at strong coupling Eq. (2) and (3) give,

$$- \frac{\beta(g)}{g} = 1 + 0(g^{-4}) \tag{4b}$$

If the theory's exact β-function smoothly interpolates between these limiting forms without the existence of a zero, then one has good evidence for the conjecture that pure gluon dynamics confines quarks with a linearly rising long-range potential. We have attempted to answer this question by calculating Eq. (2) to $0(g^{-24})$ in the hope that a strong coupling series can describe the physics of the intermediate coupling region. The exact series coefficients for the SU(3) theories in both 2 + 1 and 3 + 1 dimensions are listed in Tables 1 and 2. In more accessible, approximate form the series are, in 2 + 1 dimensions,

$$W(x) = \frac{4}{3} (1 - .0269608x^2 - .0140165x^3 - .00418557x^4 - .000864349x^5 - .0002098128x^6) \tag{5a}$$

$$- \frac{\beta(g)}{g} = 1 - .10784314x^2 - .08409927x^3 - .02476195x^4 + .00571660x^5 + .00853865x^6 \tag{5b}$$

and, more interesting, in 3 + 1 dimensions

$$W(x) = \frac{4}{3} (1 - .05392157x^2 - .02803309x^3 - .009533626x^4 - .003213169x^5 - .001824877x^6) \tag{6a}$$

$$- \frac{\beta(g)}{g} = 1 - .2156863x^2 - .1681985x^3 - .04137858x^4 + .025308657x^5 + .022764598x^6 \tag{6b}$$

These series were obtained using Rayleigh-Schrödinger perturbation theory in the variable x. Thorough tables of the Racah coefficients for SU(3) were necessary for the calculation.[4] The group theory techniques and lattice perturbation theory will be discussed in detail elsewhere. The group theory technology is quite sophisticated and we have developed many checks in this calculation by exploiting special properties and sum-rules involving the Racah coefficients.[4] In fact, these sum-rules allowed us to discover an error in our previously published x^5 coefficients.[1]

Consider the 3 + 1 dimensional theory. In Fig. 1 we plot the strong coupling series for $\beta(g)/g$ as well as the weak coupling expansion from Eq. (4a). The strong coupling series is plotted down to $g \approx 1.05$ where its high order terms begin overwhelming its low order terms. It is not meaningful to plot the series beyond this point without the use of extrapolation techniques. Two observations about Fig. 1 are in order. First, the strong coupling series tends to match onto the weak coupling series for $g \approx 1.0 - 1.1$ and the strong coupling series has no zero. These curves, therefore, suggest that the theory is characterized by three regions in coupling: $g \lesssim 1.05$ is weak coupling, $1.05 < g < 1.5$ is the crossover region, and $g \gtrsim 1.5$ is strong coupling. And second, no extrapolation methods are needed to describe the intermediate coupling region using strong coupling expansions of sufficient length.

Since the β-function of Fig. 1 does not have a zero other than at zero coupling, this calculation suggests that the theory dynamically generates a nonzero string tension T. Since the theory has no other dimensional parameters in its continuum limit, once T is fixed all other mass parameters are predictable. For example, there is the quantity Λ which sets the scale of deviations from free field behavior in the weak coupling region of the theory.[5] The weak coupling β-function predicts that Λ depends upon "a" and g as,

$$\Lambda_{sl} = \frac{1}{a}\left(\frac{16\pi^2}{11g^2}\right)^{51/121} e^{-8\pi^2/g^2} \; [1 + O(g^2)] \tag{7}$$

where the subscript "sl" indicates the use of a "spatial lattice" cutoff to regulate the theory. Since they both have dimensions of mass, $\sqrt{T}$ and Λ_{sl} must be proportional

$$\sqrt{T} = C\, \Lambda_{sl} \tag{8}$$

and it is interesting to estimate C from our strong coupling series. So, in Fig. 2 we have presented semi-log plots of T to see if the strong coupling calculations can be fit with the scaling law $\exp(-8\pi^2/g^2)$ in the weak coupling region. Actually $\sqrt{2}\, x^{-51/121}\, a^2\, T$ is plotted against $\sqrt{x}$ to accommodate the 2 loop correction to the weak coupling β function and to set the scale of the scale breaking parameter Λ_{sl} in a conventional fashion.[5] In the figure the three, four, five and six term series for T are plotted and each is fit with an exponential. The tendency of the strong coupling series to converge to a limiting exponential as the order of the calculation increases is shown in Fig. 3. For each curve in Fig. 2 we have read off a value of $C^{(i)}$, i = order of the strong coupling expansion, for T and in Fig. 3 we have plotted $C^{(i)}$ against $1/i$. We are interested in the value of $C^{(i)}$ at $1/i = 0$. An estimate can be obtained by fitting the $i = 4,5$ and 6 values with a quadratic polynomial in $1/i$,

$$C^{(i)} = C + a/i + b/i^2 \tag{9}$$

and inferring $C = 83$ as shown in the figure. An estimate of the uncertainty in this procedure is made by attempting a linear fit to $C^{(i)}$ using only the $i = 5$ and 6 term series. This procedure gives the other extrapolation in Fig. 3 and the value $C = 54$. So, we estimate finally,

$$\sqrt{T} = (69 \pm 15) \cdot \Lambda_{s1} \quad . \tag{10}$$

The large proportionality between Λ_{s1} and $\sqrt{T}$ is an artifact of lattice regularization. Each regularization procedure leads to a different Λ parameter and the relations between them have been calculated. The relation between continuum space-time momentum-space regularization Λ_{mom} and a space-time symmetric Euclidean scheme Λ_e is,[6]

$$\Lambda_{mom}/\Lambda_e = 83.5 \qquad (SU(3)) \tag{11a}$$

and the relation to the time-continuum spatial lattice used here is,[7]

$$\Lambda_{s1}/\Lambda_e = 3.012 \qquad (SU(3)) \tag{11b}$$

Now Eq. (10) can be written in a form completely free of the lattice,

$$\sqrt{T} = (2.50 \pm .50) \; \Lambda_{mom} \quad . \tag{12}$$

This is a physically reasonable result in good agreement with recent computer simulations.[3]

Eq. (10) differs from our earlier estimates of the relation between $\sqrt{T}$ and Λ_{s1} by a factor of between 2 and 3. The reason for this difference comes from the correction of the fifth order coefficients in the series Eq. (6). With the new coefficient the weak coupling edge of the crossover region has shifted to slightly larger coupling (from $g \approx .9-1.0$ to $g \approx 1.05-1.10$). Since C depends exponentially on $1/g^2$ this small change is magnified into a substantial decrease in C.

Next consider the spatial distribution of the chromo-electric flux tube extending between the static quark and anti-quark. Place the quarks along a coordinate axis. In the strong coupling regime the flux tube between them has a finite width on the order of a lattice spacing. We are interested in determining the spatial distribution of the flux in the continuum limit. It is known that in the string model, the flux wanders without bound as the quarks are separated. Suppose that the quarks are a distance L apart. Then one computes that the transverse spatial distribution of the string between them diverges as log L. This is an old result of the string model and has been discussed recently from a more modern viewpoint.[8] It has also been suggested that this changeover from narrow flux tubes to diffuse, wandering flux tubes occurs in lattice theories at a finite coupling[≠1] and that this phenomenon is a generalization of the roughening transition studied in statistical physics.[9] This suggestion raises the possibility that finite lattice computer simulations and strong coupling expansions are inadequate tools to discuss the crossover phenomenon in lattice gauge theories. Computer simulations would be unable to accommodate large scale fluctuations in the string at and below the roughening coupling g_r and therefore their estimates of the string tension could be wrong.[≠2] Strong coupling expansions would be unable to discuss the theory for $g < g_r$ because of non-analytic terms[9] in the string tension at $g = g_r$. Roughly speaking, at and below g_r one expects

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≠1 The authors are indebted to Dr. P. Hasenfratz for informative correspondence on this point.

≠2 Roughening is expected to reduce the string tension.

massless excitations to propagate along the infinitely long string even though its tension is different from zero for all $g \neq 0$. These massless excitations generate nonanalytic terms in the tension which conventional expansion methods could not approximate reliably.

We have studied roughening in Z_2, U(1) and SU(3) Hamiltonian gauge theories in 2 + 1 and 3 + 1 dimensions by several methods.[≠3] One method considers the excitation spectrum of the infinitely long string. Another, which will be discussed here because it is closely related to the string tension series, calculates by expansion methods the width of the chromo-electric flux tube directly.[8] We define this quantity as follows. Separate the quarks along the Z axis. Then the width of the tube is,

$$\frac{\langle r_\perp^2 E_Z^2 \rangle}{\langle E_Z^2 \rangle} \tag{13}$$

where E_Z^2 measures electric flux on links parallel to the unperturbed string, $r_\perp^2 = x^2 + y^2$, and the expectation value $\langle \quad \rangle$ is taken in the eigenstate of the string. A dimensionless measure of the width of the tube is given by[≠4]

$$\frac{\langle r_\perp^2 E_Z^2 \rangle}{\langle E_Z^2 \rangle} \, T \; . \tag{14}$$

The roughening transition is characterized by the divergence of this quantity. For SU(3) in 3 + 1 dimensions, we have

$$\begin{aligned} a^2 \langle r_\perp^2 E_Z^2 \rangle &= .065845401x^2 + .045184945x^3 + .024762650x^4 + .015147301x^5 \\ \langle E^2 \rangle &= \frac{4}{3}\,(1 - .044846534x^2 - .011711241x^3 - .004902915x^4) \end{aligned} \tag{15}$$

≠3 Many of the calculations were done in collaboration with Dr. J. Richardson and Dr. K. K. Sinclair.

≠4 The authors thank Professor M. Lüscher for discussions concerning width calculations.

Using the series for T we have,

$$\frac{\langle r_\perp^2 E_Z^2\rangle}{\langle E_Z^2\rangle}\cdot T = (.04655973x^{3/2})(1+.68622781x+.366997548x^2+.207494028x^3) \qquad (16)$$

and we can test for the tendency of this series to diverge by looking for the poles in its Padé approximates. The Padé approximates and the locations of their poles are,

$$\begin{aligned} &[1/1] \quad x_r = 1.88 \\ &[2/1] \quad x_r = 1.77 \\ &[1/2] \quad x_r = 1.79 \end{aligned} \qquad (17)$$

So, there is evidence of roughening at $x_r \approx 1.80 \pm .03$. Obviously this is a crude calculation since the series Eq. (16) has only four non-trivial terms. However, more sophisticated calculations of the mass spectrum of the string give $x_r \gtrsim 1.77$ in good agreement with Eq. (17).[10] An interesting feature about this result is that x_r lies <u>inside</u> the weak coupling region of the theory and does not present a barrier to strong coupling calculations of the crossover region! In the variable x the three regions of the theory as implied by Fig. 1 and refined by Eq. (10) are: weak coupling $1.44 < x < \infty$, crossover $.40 < x < 1.44$, strong coupling $x < .40$. The point is that $x_r \approx 1.80$ is greater than 1.44, the weak coupling edge of the crossover region.

Estimates of the roughening transition in the Euclidean formulation of the theory suggest that the transition occurs in the crossover region[11] or perhaps even on its strong coupling edge.[8] Since critical points are non-universal these results do not necessarily disagree with our Hamiltonian estimates although such differences are quite perplexing.

A Hamiltonian solid-on-solid model estimate in 2 + 1 dimensions also suggests, in agreement with our series calculations, that the roughening transition occurs inside the weak coupling region.[9)]

Besides being of technical interest, roughening transitions are probably also relevant to the physics of the theory's continuum limit. This topic will be developed elsewhere.[10)]

In summary, we have presented series evidence for confinement in pure gluon gauge theory, have estimated the proportionality between the scale breaking parameter Λ and the string tension and have suggested that if roughening occurs in the Hamiltonian version of the gluon gauge theory it does so in the weak coupling regime.

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Table 1

Series Coefficients for the String Tension

2 + 1 Dimensions

t_0	$4/3$
t_1	0
t_2	$-\frac{11}{306}$
t_3	$-\frac{61}{3264}$
t_4	$-\frac{647421073157710219}{116009444617639372800}$
t_5	$-\frac{1037670685176149777004438 1}{900391836473858973404160000}$
t_6	$-\frac{618968838776800613312540645449998271988535582347430995 23}{221257574537828516967753337513669558600439838693961728000000}$

Table 2

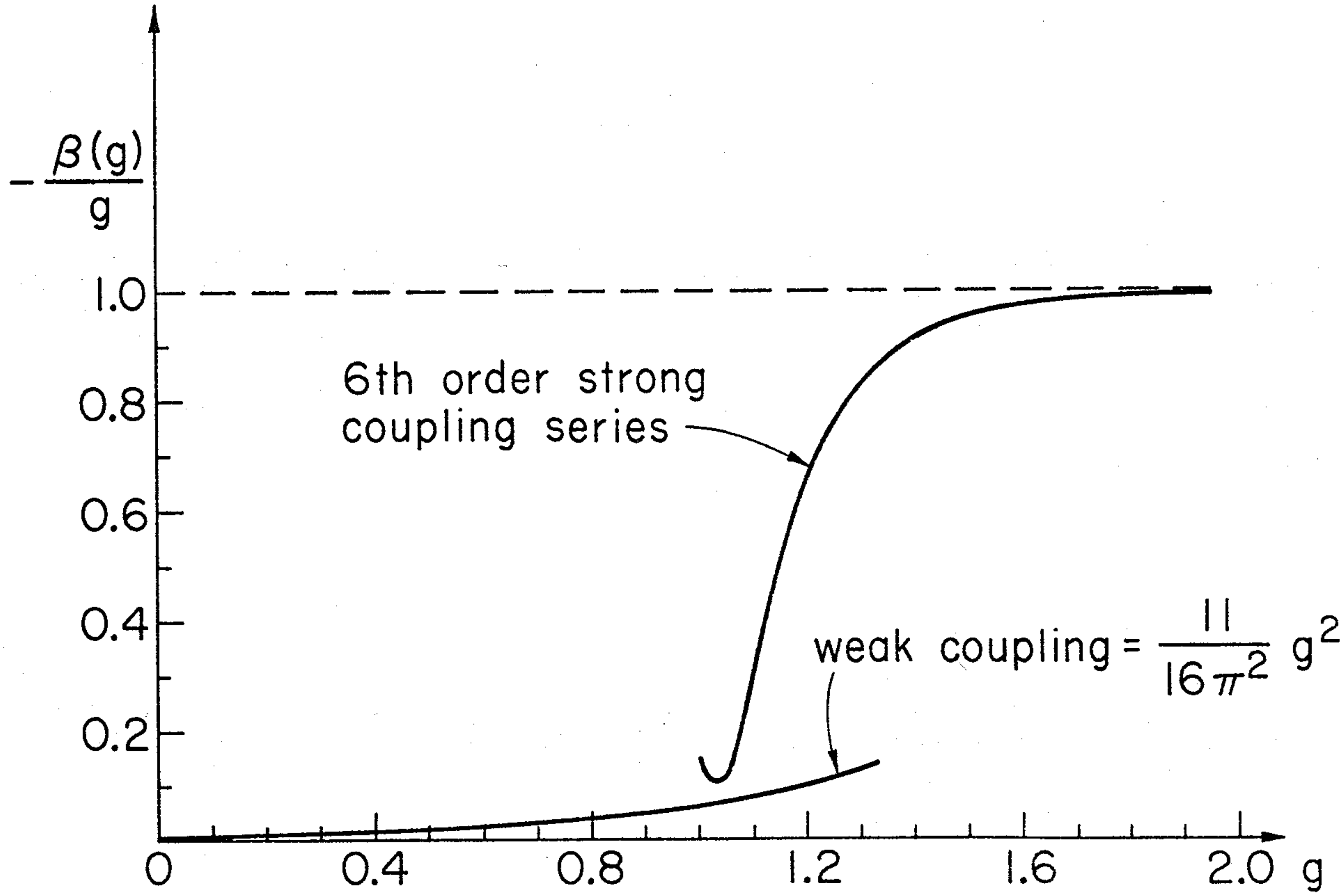
Series Coefficients for the String Tension

3 + 1 Dimensions

t_0	$4/3$
t_1	0
t_2	$-\frac{11}{153}$
t_3	$-\frac{61}{1632}$
t_4	$-\frac{737327120374220449}{5800472230881968640 0}$
t_5	$-\frac{1377672221891827359503 09}{3215685130263782047872 0000}$
t_6	$-\frac{131306616610341907729359593488164446498007144107500 15999}{539652620823971992604276432960169655123023996814540 8000000}$

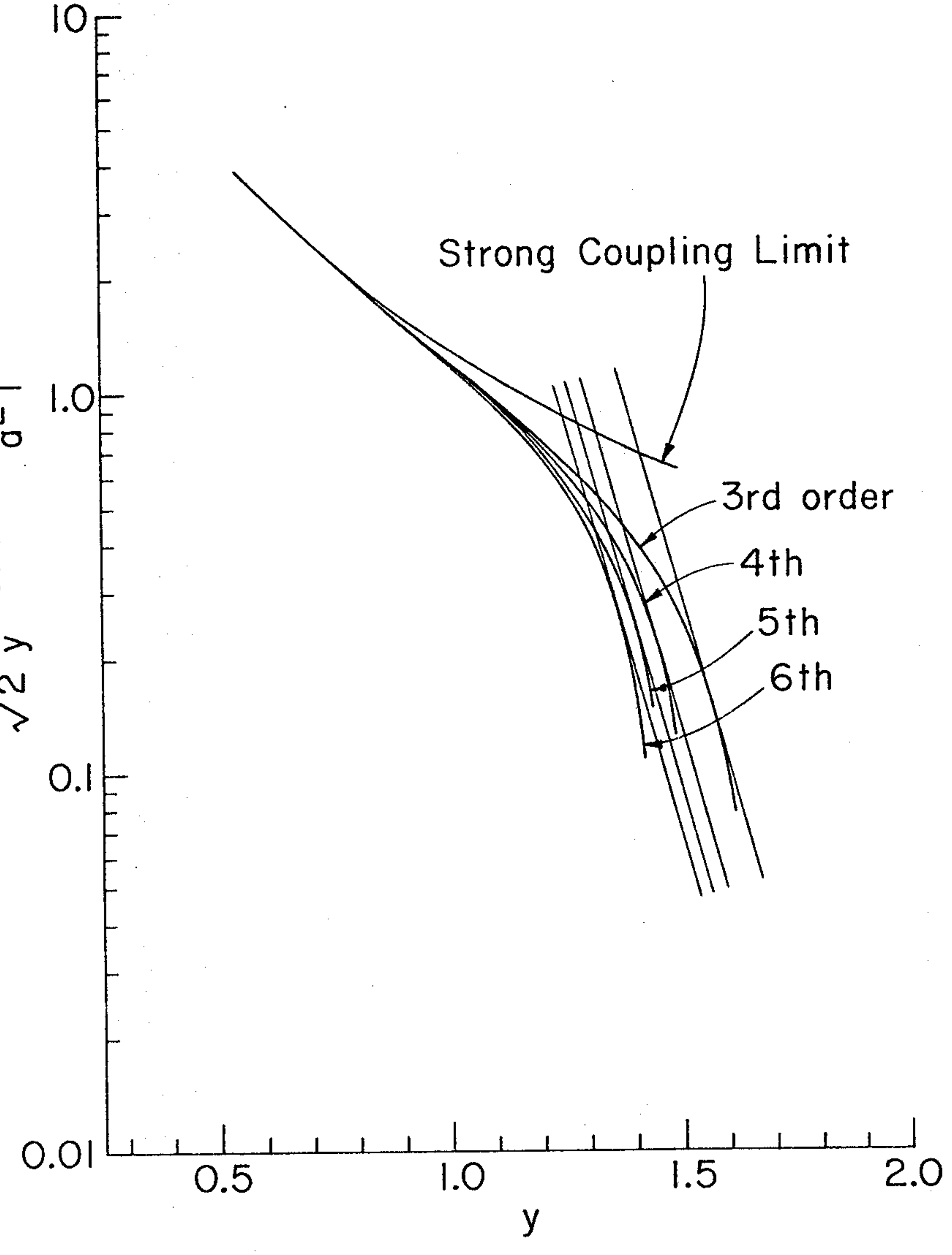
Figure Captions

1. Weak and strong coupling expansions for the β-function of Hamiltonian SU(3) lattice gauge theory in 3 + 1 dimension.
2. Strong coupling expansions of the string tension. The strong coupling series of 3, 4, 5 and 6 terms are plotted and their weak coupling ends are fit with the exponential of asymptotic freedom. Here $y = \sqrt{2}/g^2$.
3. $C^{(i)}$ vs. $1/i$. The dash-dot curve is a quadratic fit to the $i = 4,5$ and 6 points. The dash-dash line is a linear fit to the $i = 5$ and 6 points.



$-\frac{\beta(g)}{g}$
1.0
0.8
0.6
0.4
0.2
0
0.4
0.8
1.2
1.6
2.0
g
6th order strong coupling series
weak coupling = $\frac{11}{16\pi^2} g^2$

Fig. 1



Strong Coupling Limit
3rd order
4th
5th
6th
10
1.0
0.1
0.01
0.5
1.0
1.5
2.0
y
$\sqrt{2}\, y^{-102/121}\, a^2 T$

Fig. 2

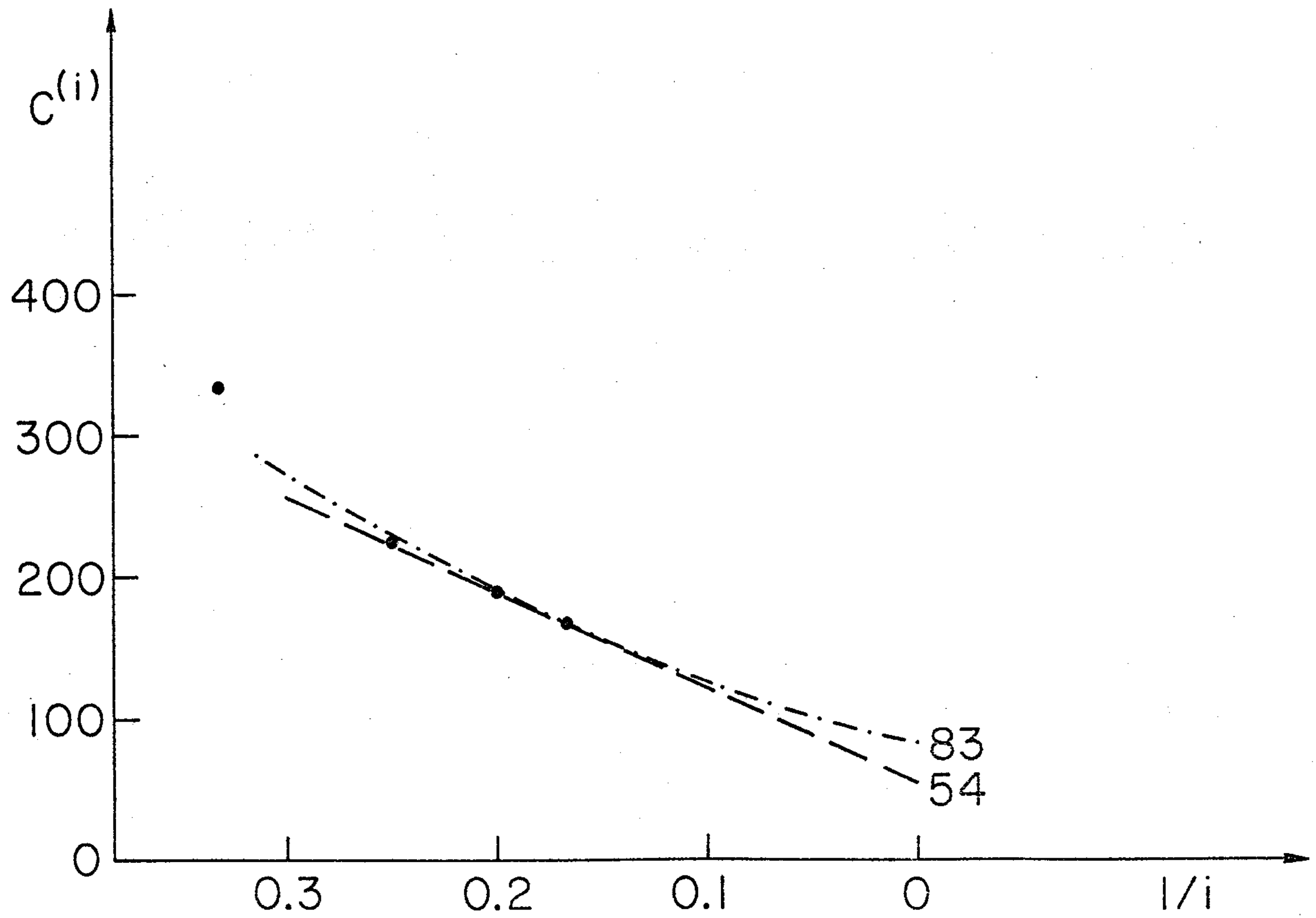


Fig. 3

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